

BTC 5-Minute Strategy Research Report

Candidate strategies benchmarked against the existing **Sway model**. Generated 2026-06-28 04:43:11 UTC.

Executive Summary

Trained on **600** historical markets, tested out-of-sample on **250** more-recent markets (test base rate 53.2% UP). Best by accuracy: **Ensemble** (89.3%). The **Sway baseline is the weakest model** (77.4% accuracy) — worse than simply trusting the live market price (89.0%).

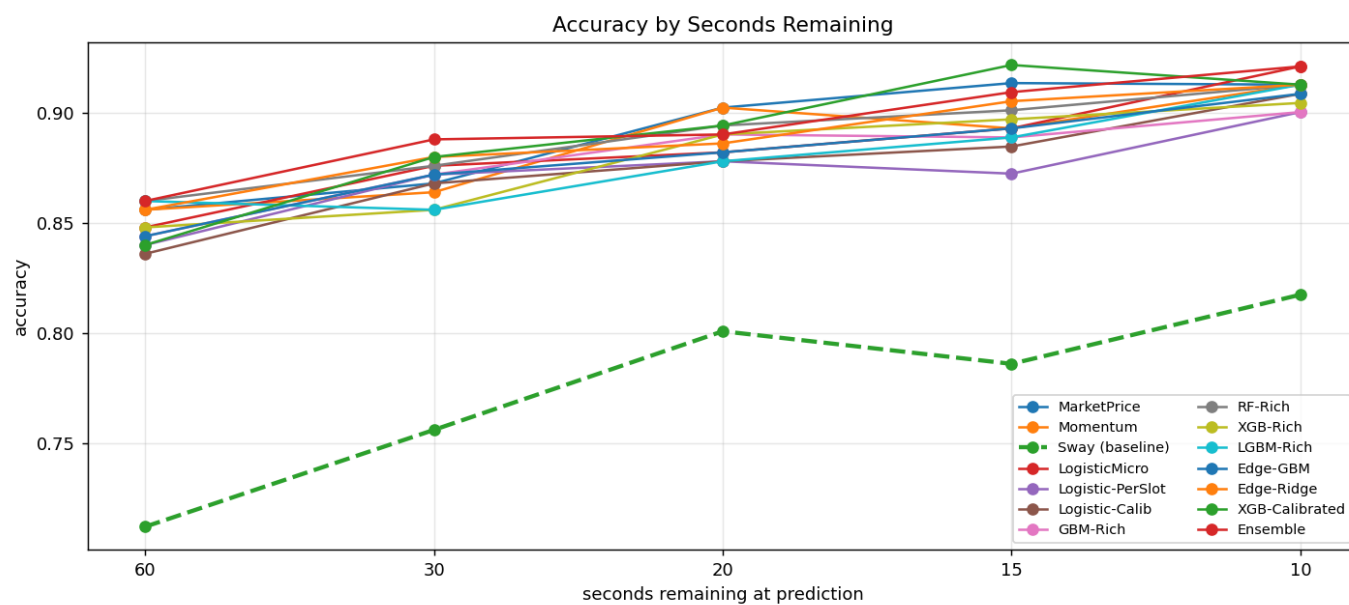
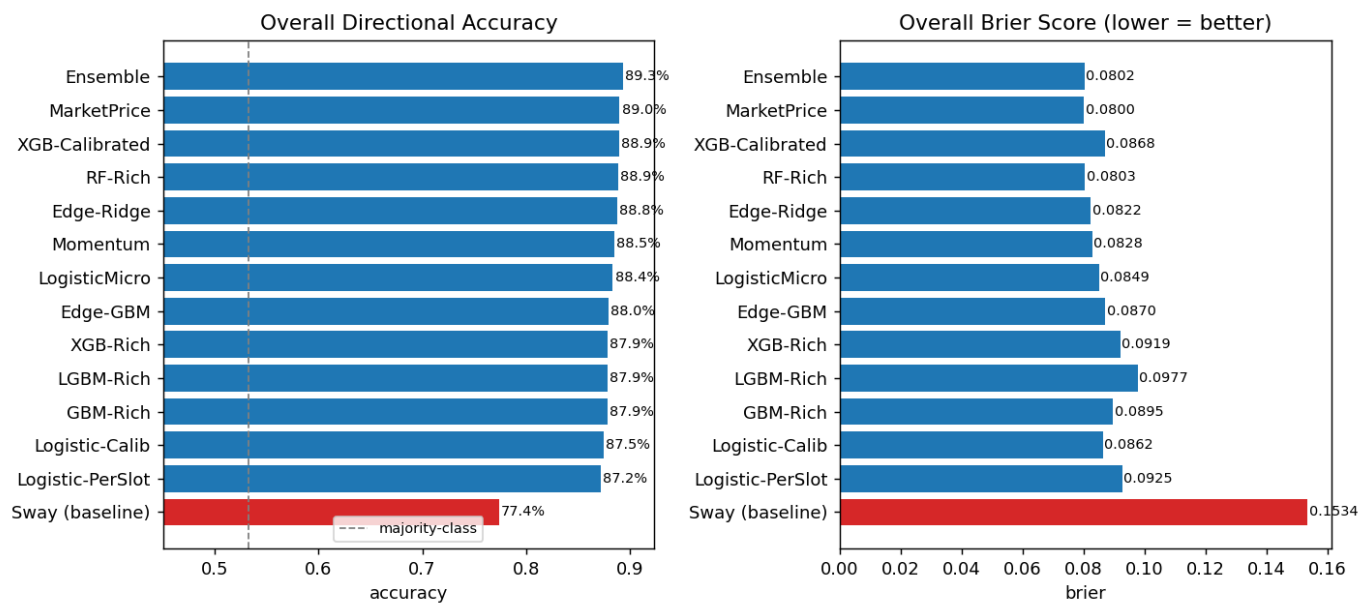
Headline result. Re-tested on a second independent window (see Cross-Window Robustness), single-window trading ROI proves to be mostly noise — most strategies flip between profit and loss. The only strategy profitable on **both** windows is **LogisticMicro** (+3.3% / +7.8%), making it the recommended candidate for live trading. Statistical ranking, by contrast, is stable: Sway is consistently last.

Overall metrics (out-of-sample)

Strategy	Accuracy	Brier	LogLoss	Bets	ROI	P&L \$	WinRate	MaxDD\$
Ensemble	89.3%	0.0802	0.2579	426	-9.3%	-39.7	54.0%	62.7
MarketPrice	89.0%	0.0800	0.2547	0	+0.0%	+0.0	0.0%	0.0
XGB-Calibrated	88.9%	0.0868	0.2867	601	-8.4%	-50.4	47.6%	132.6
RF-Rich	88.9%	0.0803	0.2569	476	-2.1%	-10.1	71.0%	44.4
Edge-Ridge	88.8%	0.0822	0.3140	620	-29.0%	-179.5	49.0%	182.2
Momentum	88.5%	0.0828	0.3403	370	-8.9%	-33.1	63.8%	46.7
LogisticMicro	88.4%	0.0849	0.2768	548	+3.3%	+17.9	70.3%	33.7
Edge-GBM	88.0%	0.0870	0.3211	540	-13.9%	-75.0	54.6%	84.9
GBM-Rich	87.9%	0.0895	0.2928	603	-7.8%	-47.1	61.5%	77.4
XGB-Rich	87.9%	0.0919	0.3069	613	-0.9%	-5.5	69.2%	49.9
LGBM-Rich	87.9%	0.0977	0.3400	632	-2.0%	-12.5	70.4%	36.2
Logistic-Calib	87.5%	0.0862	0.3023	538	-12.4%	-66.8	50.7%	80.6
Logistic-PerSlot	87.2%	0.0925	0.3193	593	-1.4%	-8.4	67.5%	33.2
Sway (baseline)	77.4%	0.1534	0.5624	890	-27.3%	-242.9	28.2%	288.5

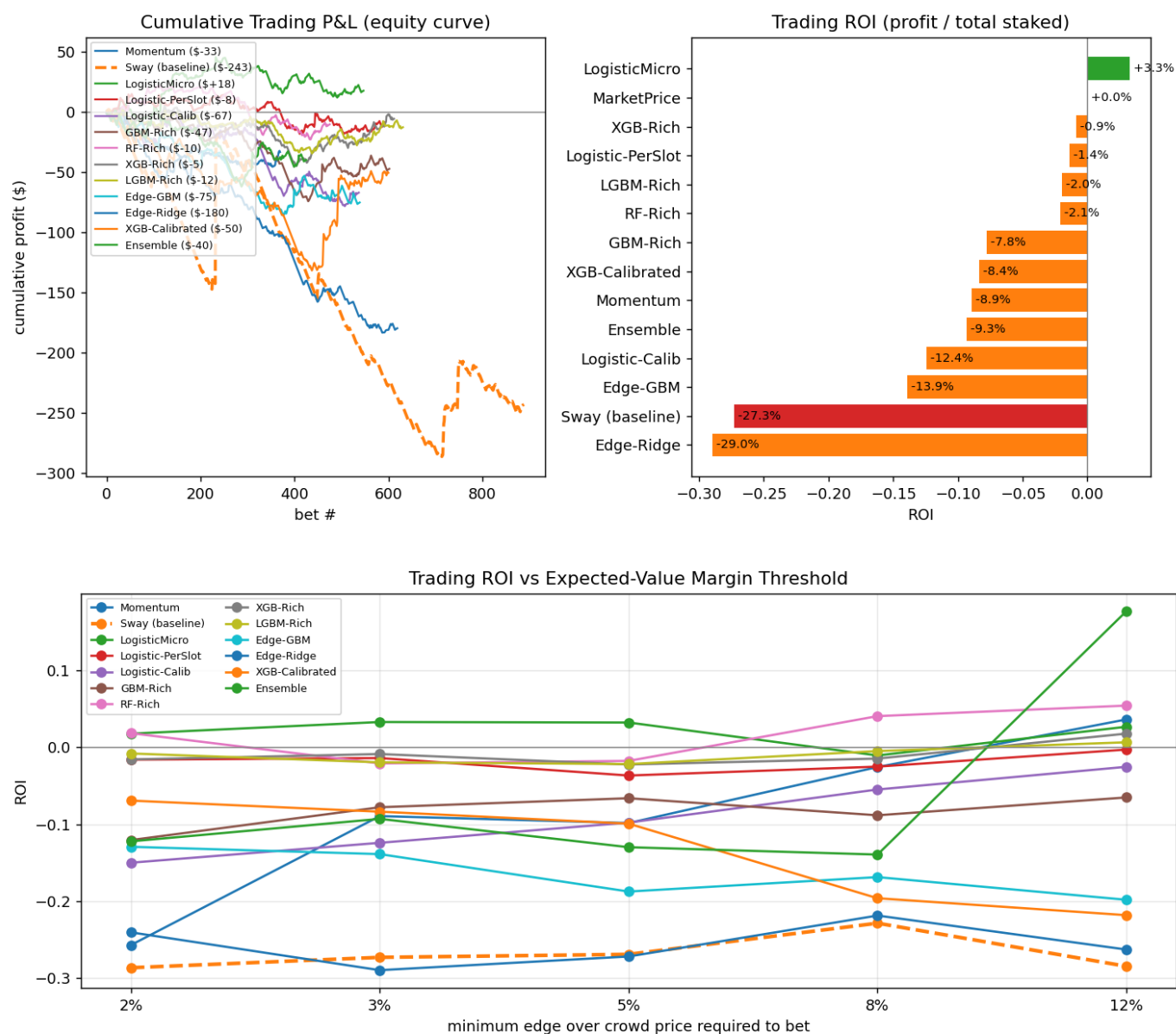
Sway baseline highlighted in red; best strategy in bold. Trading simulation bets \$1 whenever a model's probability diverges from the live market price by > 3% (positive expected value), including Polymarket fees.

Accuracy & Calibration



Trading Performance

The sway model ignores the absolute market price, so its statistical edge does not always convert into trading profit. The equity curves below show realised P&L when each model only bets on positive-EV divergences from the crowd price.



Cross-Window Robustness

To separate genuine edge from luck, every strategy is re-evaluated on a second, fully independent and time-disjoint window of **300** older markets, in a different regime (45.7% UP vs 53.2% in the recent test window).

Two clear conclusions: (1) Statistical quality is robust — the Sway baseline is the least accurate model in *both* windows by a wide margin. (2) Single-window trading ROI is largely noise: most strategies flip sign between windows (e.g. the Ensemble swings from -9% to +34%, Edge-GBM from -14% to +25%, even Sway from -27% to +8%). The only strategy with **positive ROI on both** independent windows is **LogisticMicro** — the lone candidate showing a repeatable trading edge.



Methodology

All strategies use only trade data available up to the prediction time (no look-ahead). Predictions are made at 60, 30, 20, 15 and 10 seconds remaining. The Sway baseline is a faithful reproduction of the repo's per-slot GradientBoosting model on the 29 channel-sway features. Candidate models add the absolute price level, VWAP, multi-horizon momentum/volatility and order-flow imbalance features. Metrics: directional accuracy, Brier score and log-loss (probability quality), plus a fee-aware Polymarket P&L simulation.